

Beckett Frey

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EDUCATION

UW Milwaukee

B.S. Computer Science

Milwaukee, WI, USA

January 2023 - May 2025

- Dean's Honor List, Fall 2024 & Spring 2025

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, Bash, SQL, R

Tools and Frameworks: Django, React, PyQt6, LangGraph, Node.js

Cloud and Infra: GitHub Actions, Docker, Modal, AWS

Machine Learning: Speech (Whisper, Pyannote, Forced Alignment), Vision (YOLO, Synthetic Data)

WORK EXPERIENCE

Research Intern

Waisman Center, WI, USA

September 2025 - Present

- Build and ship a cross-platform desktop application for phonetic research, reducing onboarding friction and improving metadata tracking for reproducible workflows.
- Develop tooling supporting machine learning workflows for automated pronunciation assessment in children with speech sound disorders.
- Build CI/CD pipelines with GitHub Actions and Docker across multiple repositories, improving reliability and reducing maintenance overhead for researchers.
- Collaborate across labs to support the ongoing development of legacy Django infrastructure, focusing on modernization and knowledge continuity.

Software Consultant & Engineer (Contract)

Independent, Madison, WI, USA

June 2026 - Present

- Performed application security review of a production SaaS product and delivered prioritized written recommendations to the client.
- Coordinated an independent third-party security assessment on the client's behalf, from firm evaluation through scoping and engagement.
- Built out separate development and staging infrastructure to move code changes out of production workflows.
- Advise a prospective startup founder on technical feasibility, architecture, and MVP scoping.

Software Engineer (Volunteer)

Environmental Data & Governance Initiative, Remote, USA

January 2026 - Present

- Resolve bugs and deliver features for the web-monitoring project in response to analyst feedback.
- Improve CI with ruff and git hooks to gate low-quality open-source contributions before review.

Junior ML Engineer

Hyperchess.ai, Remote, USA

December 2024 - September 2025

- Built a synthetic data pipeline and a 25-model YOLO mixture-of-experts for piece detection, reaching 99% mAP@0.5.
- Deployed inference workloads with Modal serverless GPU, drastically improving the speed of iteration.

PROJECTS

- **Phonetic Research Application**, Cross-platform desktop app that unifies AI/ML tools (MFA, Wav2TextGrid, Faster-Whisper, PyLLRComputer) behind one interface, letting Speech Language Pathology researchers build and run phonetic analysis pipelines in one place. To be presented at Interspeech 2026. [Try it!](#)
- **AI Document Generation**, Python CLI and library that fills markdown templates with LLM-generated content under strict length and structure constraints, using batched context-aware generation and a LangGraph morphing loop to enforce character bounds. [GitHub](#)
- **Data Security Tooling**, Published PyPI tool that scans git repositories for PII and other identifier patterns. Used to safely open-source several lab repos and caught one real PII exposure in legacy code. [Try it!](#)

EXTRACURRICULAR ACTIVITIES

- **Open Source Contributor**, Github - Ongoing
- **Inpatient Unit Support**, UW Health - Ongoing
- **Triathlete**, IRONMAN Wisconsin - September 2023